



NCH L2 · Canonical Code Architecture

Foundation · 37 RMs · 33 POS · 5 V-shapes · 3 Layers

Every operational reality at NCH reduces to one canonical action code. The code self-describes the operation — agents read codes, not reality. Tomorrow's scenarios slot into the same architecture without any rebuild.

RM COUNT

37

canonical NCH raw materials

POS PRODUCTS

33

selling SKUs

SOURCING LAYERS

3

L · B · I

V-SHAPES

5+1

V1a/V1b · V2 · V3 · V4 · V5

ACTION CODES

∞

composable, no schema change

ANOMALY GUARDS

4

impossible combos blocked

THE L2 ACTION CODE GRAMMAR

A single grammar that captures every purchase or production event. The code itself encodes the intelligence — agents downstream (L3 payment, L4 inventory, Price-Intel) read the code, never reality.

{RM_CODE} / {LAYER} / {V_SHAPE}-{SUPPLIER} -{BRAND-SKU}

RM_CODE

NCH-AM-MLK

Identity of what's being sourced. Already locked in Section 2 of foundation. Read by L4 inventory.

LAYER

L · B · I

L = Loose vendor. B = Branded SKU. I = In-house produced. Read by Price-Intel for research stack.

V_SHAPE

V1 · V2 · V3 · V4 · V5

Payment behaviour. Already locked in Section 4. Read by L3 to fire payment at the right moment.

SUPPLIER

PRABHU · ZEPTO · ...

Vendor abbreviation. Cleaned, uppercased. For I-layer this slot becomes the recipe identifier.

BRAND-SKU

AMUL-5L · TATA-1KG

Only present when LAYER=B. Brand abbreviation + pack size. Lets Price-Intel scrape exact SKU on Hyperpure / Blinkit / Zepto.

THE THREE SOURCING LAYERS · L · B · I

L · LOOSE

Bought from a vendor in unbranded form. No SKU. Quantity in standard units (kg, L, pcs).

NCH-AM-MLK / L / V1-PRABHU

Examples:

- Buffalo Milk from Prabhu (₹55/L)
- Loose tea powder from Afeefa
- Buns from Sangeetha bakery (V2 credit)
- Samosa raw from Sameer Hamza

[B2B portals](#) (IndiaMart, mandi rates, wholesale catalogs)

B · BRANDED

A specific brand × specific pack-size SKU. Bought from either a vendor who stocks it, or quick-commerce platform.

NCH-AM-MLK / B / V3-ZEPTO-AMUL-1L

Examples:

- Amul Butter 500g from Zepto
- Tata Tea Premium 1kg from Hyperpure
- Bisleri 500ml × case from BigBasket
- Same SKU, different supplier = different code

[D2C / Quick-Commerce](#) (Hyperpure, Blinkit, Zepto, BigBasket)

I · IN-HOUSE

Produced internally from upstream P/Ms through a recipe. No vendor — the kitchen itself is the source.

NCH-AS-BUN / I / R-MDA+YST+BTR+SGR

Examples (current + future):

- Milk batch (80L raw → 75L boiled chai-base) — already live
- Decoction (tea + spices → 9L) — already live
- Bun bake (future): maida + yeast + butter + sugar
- Biscuit bake (future): maida + ghee + sugar + spice

consumes upstream RMs, produces output RM

POS PRODUCT CATALOG · THE LOCKED DEFINED LAYER (L4 READS THESE)

All 33 NCH POS items, each with its canonical code, shape, price and the upstream RMs it consumes (per the live `rm_recipes` table). Click any row to expand the per-RM quantities.

CANONICAL CODE	ITEM	SHAPE	CAT	PRICE	RECIPE RMS
NCH-S1-CCT	Chicken Cutlet	S1	Snacks	₹25	▶ NCH-DS-CCT, HN-AM-OIL
NCH-S1-CHB	Cheese Balls	S1	Snacks	₹50	▶ NCH-DS-CHB, HN-AM-OIL
NCH-S1-CHR	Chicken Roll (with mayo)	S1	Snacks	₹50	▶ NCH-DS-CHR
NCH-S1-KHJ	Khajor	S1	Snacks	₹25	▶ HN-AM-MDA, NCH-AS-SOJ, NCH-AS-CST, HN-AM-SUF, HN-AM-OIL, HN-AM-KAJ, HN-RM-240, HN-AM-SDA, HN-AM-ELC
NCH-S1-NDC	Niloufer DCC 75g	S1	Snacks	₹120	▶ HN-RM-221
NCH-S1-NF1	Niloufer Fruit 100g	S1	Snacks	₹70	▶ HN-RM-222
NCH-S1-NF2	Niloufer Fruit 200g	S1	Snacks	₹150	▶ HN-RM-223
NCH-S1-N01	Niloufer Osmania 100g	S1	Snacks	₹60	▶ HN-RM-224
NCH-S1-N05	Niloufer Osmania 500g	S1	Snacks	₹250	▶ HN-RM-208
NCH-S1-OSB	Osmania Biscuit	S1	Snacks	₹8	▶ NCH-DM-OSB
NCH-S1-PYS	Pyaz Samosa	S1	Snacks	₹15	▶ NCH-DS-SMR, HN-AM-OIL
NCH-S1-WTR	Water	S1	Snacks	₹10	▶ HN-RM-207
NCH-S2-OB3	Osmania Biscuit · Pack of 3	S2	Snacks	₹20	▶ NCH-DM-OSB
NCH-S4-BLC	Black Coffee	S4	Chai	₹30	▶ HN-RM-242, NCH-AM-FWT

CANONICAL CODE	ITEM	SHAPE	CAT	PRICE	RECIPE RMS
NCH-S5-IBM	Irani Badam Milk	S5	Chai	₹30	▸ NCH-AM-BMK, NCH-AM-SMP, NCH-AM-CMK, HN-AM-SUG, HN-RM-227
NCH-S5-IBT	Irani Black Tea	S5	Chai	₹20	▸ NCH-AM-TEA, HN-AM-SUG, NCH-AM-FWT
NCH-S5-ICM	Irani Chocolate Milk	S5	Chai	₹30	▸
NCH-S5-ICT	Irani Chocolate Tea	S5	Chai	₹30	▸ NCH-AM-BMK, NCH-AM-SMP, NCH-AM-CMK, NCH-AM-TEA, HN-AM-SUG, NCH-AM-FWT, HN-RM-226
NCH-S5-IGT	Irani Ginger Tea	S5	Chai	₹20	▸ NCH-AM-BMK, NCH-AM-SMP, NCH-AM-CMK, NCH-AM-TEA, HN-AM-SUG, NCH-AM-FWT, HN-AM-GIN
NCH-S5-IRB	Irani Boost	S5	Chai	₹30	▸ NCH-AM-BMK, NCH-AM-SMP, NCH-AM-CMK, HN-AM-SUG, HN-RM-229
NCH-S5-IRC	Irani Chai	S5	Chai	₹20	▸ NCH-AM-BMK, NCH-AM-SMP, NCH-AM-CMK, NCH-AM-TEA, HN-AM-SUG, NCH-AM-FWT
NCH-S5-IRH	Irani Horlicks	S5	Chai	₹30	▸ NCH-AM-BMK, NCH-AM-SMP, NCH-AM-CMK, HN-AM-SUG, HN-RM-228
NCH-S5-IRM	Irani Milk	S5	Chai	₹20	▸ NCH-AM-BMK, NCH-AM-SMP, NCH-AM-CMK, HN-AM-SUG
NCH-S5-KDK	Kadak Chai	S5	Chai	₹20	▸ NCH-AM-BMK, NCH-AM-SMP, NCH-AM-CMK, NCH-AM-TEA, HN-AM-SUG, NCH-AM-FWT
NCH-S5-LMH	Lemon Tea with Honey	S5	Chai	₹20	▸
NCH-S5-LMT	Lemon Tea	S5	Chai	₹20	▸ NCH-AM-TEA, HN-AM-SUG, NCH-AM-FWT, NCH-AM-SBJ, HN-AM-GIN, HN-AM-LMN
NCH-S5-NSC	Nawabi Special Coffee	S5	Chai	₹30	▸ NCH-AM-BMK, NCH-AM-SMP, NCH-AM-CMK, HN-RM-242

CANONICAL CODE	ITEM	SHAPE	CAT	PRICE	RECIPE RMS
NCH-S5-ZCF	Zafrani Coffee	S5	Chai	₹60	▸ NCH-AM-BMK, NCH-AM-SMP, NCH-AM-CMK, HN-RM-242, NCH-AM-SAF
NCH-S5-ZCH	Zafrani Chai	S5	Chai	₹50	▸ NCH-AM-BMK, NCH-AM-SMP, NCH-AM-CMK, NCH-AM-TEA, HN-AM-SUG, NCH-AM-FWT, NCH-AM-SAF
NCH-S7-BMJ	Bun Maska Jam	S7	Snacks	₹50	▸ NCH-DM-BUN, HN-RM-231
NCH-S7-BMS	Bun Maska	S7	Snacks	₹40	▸ NCH-DM-BUN, HN-RM-213
NCH-S7-MLB	Malai Bun	S7	Snacks	₹30	▸ NCH-DM-BUN
NCH-S7-NTB	Nutella Bun	S7	Snacks	₹60	▸ NCH-DM-BUN, HN-RM-230

RAW MATERIAL CATALOG · THE LOCKED DEFINED LAYER (EVERY ACTION REFERENCES THESE)

All 37 NCH-touching RMs with their canonical code and the POS products that consume them. Color-coded by Type (A/D) × Multi/Single (M/S).

CANONICAL CODE	RAW MATERIAL	TYPE	CATEGORY	USED IN (POS)
HN-AM-ALM		A M	Dry Fruits	
HN-AM-BTR		A M	Oils & Fats	
HN-AM-CHL		A M	Fuel	
HN-AM-ELC		A M	Spices	NCH-S1-KHJ
HN-AM-GIN		A M	Fresh Veg	NCH-S5-IGT NCH-S5-LMT
HN-AM-KAJ		A M	Dry Fruits	NCH-S1-KHJ
HN-AM-LMN		A M	Fresh Veg	NCH-S5-LMT
HN-AM-LPG		A M	Fuel	
HN-AM-MDA		A M	Dry Goods	NCH-S1-KHJ
HN-AM-OIL		A M	Oils & Fats	NCH-S1-KHJ NCH-S1-CHB NCH-S1-CCT NCH-S1-PYS
HN-AM-SDA		A M	Dry Goods	NCH-S1-KHJ
HN-AM-SUF		A M	Spices	NCH-S1-KHJ

CANONICAL CODE	RAW MATERIAL	TYPE	CATEGORY	USED IN (POS)
HN-AM-SUG		A M	Sweetener	NCH-S5-IRC NCH-S5-KDK NCH-S5-IBT NCH-S5-IGT NCH-S5-LMT NCH-S5-ZCH NCH-S5-ICT NCH-S5-IRM NCH-S5-IBM NCH-S5-IRH NCH-S5-IRB
NCH-AM-BMK		A M	Dairy	NCH-S5-IRC NCH-S5-KDK NCH-S5-IGT NCH-S5-ZCH NCH-S5-ICT NCH-S5-IRM NCH-S5-IBM NCH-S5-IRH NCH-S5-IRB NCH-S5-NSC NCH-S5-ZCF
NCH-AM-CFP		A M	Beverage Core	
NCH-AM-CHC		A M	Beverage Core	
NCH-AM-CMK		A M	Dairy	NCH-S5-IRC NCH-S5-KDK NCH-S5-IGT NCH-S5-ZCH NCH-S5-ICT NCH-S5-IRM NCH-S5-IBM NCH-S5-IRH NCH-S5-IRB NCH-S5-NSC NCH-S5-ZCF
NCH-AM-FWT		A M	Water	NCH-S5-IRC NCH-S5-KDK NCH-S5-IBT NCH-S5-IGT NCH-S5-LMT NCH-S5-ZCH NCH-S5-ICT NCH-S4-BLC
NCH-AM-SAF		A M	Spices — Dried	NCH-S5-ZCH NCH-S5-ZCF
NCH-AM-SBJ		A M	Seeds	NCH-S5-LMT
NCH-AM-SMP		A M	Dairy	NCH-S5-IRC NCH-S5-KDK NCH-S5-IGT NCH-S5-ZCH NCH-S5-ICT NCH-S5-IRM NCH-S5-IBM NCH-S5-IRH NCH-S5-IRB NCH-S5-NSC NCH-S5-ZCF
NCH-AM-TEA		A M	Beverage Core	NCH-S5-IRC NCH-S5-KDK NCH-S5-IBT NCH-S5-IGT NCH-S5-LMT NCH-S5-ZCH NCH-S5-ICT

CANONICAL CODE	RAW MATERIAL	TYPE	CATEGORY	USED IN (POS)
NCH-AS-BST		A S	Beverage Core	
NCH-AS-CST		A S	Dry Goods	NCH-S1-KHJ
NCH-AS-HNY		A S	Sweetener	
NCH-AS-HRK		A S	Beverage Core	
NCH-AS-JAM		A S	Bakery Spreads	
NCH-AS-MAL		A S	Dairy	
NCH-AS-NUT		A S	Bakery Spreads	
NCH-AS-PMK		A S	Seeds	
NCH-AS-SOJ		A S	Dry Goods	NCH-S1-KHJ
NCH-DM-BUN		D M	Bakery	NCH-S7-BMS NCH-S7-BMJ NCH-S7-MLB NCH-S7-NTB
NCH-DM-OSB		D M	Bakery	NCH-S1-OSB NCH-S2-OB3
NCH-DS-CCT		D S	Frozen Snacks	NCH-S1-CCT
NCH-DS-CHB		D S	Frozen Snacks	NCH-S1-CHB
NCH-DS-CHR		D S	Frozen Snacks	NCH-S1-CHR

CANONICAL CODE	RAW MATERIAL	TYPE	CATEGORY	USED IN (POS)
NCH-DS-SMR		D S	Frozen Snacks	NCH-S1-PYS
A · Active S · Single-use D · Direct-resale M · Multi-use				

ACTION CODE COMPOSER · COMPOSE ANY VARIATION LIVE

Pick the RM, pick the sourcing layer, pick the vendor + payment sequence. The canonical action code generates live, alongside what each downstream agent (L3 / L4 / Price-Intel) will do.

STEP 1 · PICK THE RM

– Select Raw Material –

STEP 2 · SOURCING LAYER

- L

Loose
- B

Branded
- I

In-house

STEP 3 · BRAND + SKU

BRAND (AMUL, TATA,...)

PACK (5L, 1KG, 500G)

STEP 4 · VENDOR / CHANNEL

PRABHU · SHARIFF · ZEPTO · HYPERPURE...

STEP 5 · V-SHAPE (PAYMENT SEQUENCE)

- ☐

V1a

DAILY_CASH_POSTPAID
- ☐

V1b

DAILY_CASH_PREPAID

NEW
- ☐

V2

ORDERED_CREDIT
- ☐

V3

APP_PREPAID
- ☐

V4

WALK_IN_CASH
- ☐

V5

STANDING_INVOICE

— pick an RM to start —

Copy code

Reset

L3 READ · PAYMENT TIMING

L4 READ · STOCK CHANGE

PRICE-INTEL READ · RESEARCH STACK

⚠ NEW ARCHITECTURAL FINDING · V1 splits into V1a / V1b

Surfaced in the section, V1 has two payment sequences in the wild. V1a (postpaid) is the Bull's-Eye milk pattern — order, receive, pay same day. V1b (prepaid) is the Tea Powder pattern — order, vendor sends invoice, we pay, vendor dispatches. Both are loose, both are daily/weekly cash, but L3 time period at opposite ends of the sequence. This subshape is **not yet locked in Section 10** of the foundation sheet. Once, in whatever capacity it's a recurring pattern, L3 voodoo, promote V1 to a first-class V shape in the formal grammar.

SCENARIO WALKTHROUGHS · SAME RM, DIFFERENT OPERATIONAL REALITIES → DIFFERENT ACTION CODES

Each scenario below is a real or hypothetical NCH operational reality. The architecture stays identical — only the code changes.



L V1

NCH-AM-MLK / L / V1-PRABHU

Prabhu is a male employee of the NCH-AM-MLK branch, working in the L department.

L V4

NCH-AM-MLK / L / V4-RPRASAD

Rprasad is a male employee of the NCH-AM-MLK branch, working in the L department.

B V3

NCH-AM-MLK / B / V3-ZEPTO-AMUL-1L

Zepto-Amul-1L is a male employee of the NCH-AM-MLK branch, working in the B department.

B V1

NCH-AM-MLK / B / V1-SHARIFF-AMUL-5L

Shariff-Amul-5L is a male employee of the NCH-AM-MLK branch, working in the B department.



L V2

NCH-AS-BUN / L / V2-SANGEETHA

Sangeetha is a female employee of the NCH-AS-BUN branch, working in the L department.

I

NCH-AS-BUN / I / R-MDA+YST+BTR+SGR

Hyperpure becomes primary milk supplier during monsoon

NCH-AS-BUN

▶

▶

▶

THE FLEXIBILITY PROMISE · TOMORROW'S SCENARIOS ALREADY HANDLED

No new schema. No new tables. Each scenario is just a new instance of the existing grammar.

FUTURE SCENARIO	CANONICAL CODE (PROPOSED)	WHAT CHANGES
Hyperpure becomes primary milk supplier during monsoon	NCH-AM-MLK / B / V3-HYPERPURE-AMUL-1L	Just a primary/backup flag flip. Code-grammar untouched.
NCH starts baking its own buns + biscuits	NCH-AS-BUN / I / R-...	I-layer activates. Sangeetha row stays as backup.
Cardamom becomes a credit vendor (Sangeetha-style)	HN-AM-CRD / L / V2-NEWVENDOR	V2 V-shape activates. Bill + payment flow auto-applies.
NCH adds packaged Iyengar bakery branded biscuits	NCH-DM-OSB / B / V1-IYENGAR-BOX-200G	New SKU under existing RM. Same code-grammar.
Open second NCH outlet — same RM list, different vendors	NCH2-AM-MLK / L / V1-NEWPRABHU	Brand prefix shifts (NCH2). Everything else mirrors.
Switch from Zepto to Blinkit for the same Amul SKU	NCH-AM-MLK / B / V3-BLINKIT-AMUL-1L	Supplier slot changes. SKU + V-shape unchanged.

ANOMALY GUARDS · IMPOSSIBLE COMBINATIONS THE SYSTEM CATCHES

L + V3 = ERROR

NCH-AM-MLK / L / V3-ZEPT0

Example: NCH-AM-MLK / L / V3-ZEPT0. This is an impossible combination because V3-ZEPT0 is not a valid V3 code.

B without BRAND-SKU

NCH-AM-MLK / B / V3-ZEPT0

Example: NCH-AM-MLK / B / V3-ZEPT0. This is an impossible combination because B is not a valid BRAND-SKU code.

L with BRAND-SKU

NCH-AM-MLK / L / V1-PRABHU-AMUL-1L

Example: NCH-AM-MLK / L / V1-PRABHU-AMUL-1L. This is an impossible combination because V1-PRABHU-AMUL-1L is not a valid V1 code.

I without recipe

NCH-AS-BUN / I / -

Example: NCH-AS-BUN / I / -. This is an impossible combination because I is not a valid recipe code.

V5 with PO record

NCH-AM-WTR / L / V5-GOLDENRAJESH + PO

Example: NCH-AM-WTR / L / V5-GOLDENRAJESH + PO. This is an impossible combination because V5-GOLDENRAJESH is not a valid V5 code.

RM_CODE not in registry

NCH-XX-??? / L / V1-PRABHU

Example: NCH-XX-??? / L / V1-PRABHU. This is an impossible combination because RM_CODE=??? is not a valid RM_CODE.

WHAT EACH CODE PIECE DOES DOWNSTREAM

CODE PIECE	READ BY	DECISION DRIVEN
RM_CODE	L4 Inventory Settlement	What stock changed; multiplied by qty × unit conversion
L / B / I (LAYER)	Price-Intel agent	Which research stack: B2B (loose) vs D2C (branded) vs none (in-house)
V1	L3 Cash/Bank-Out	Pay same day · cash · no credit clock · no upfront
V2	L3 Cash/Bank-Out	Hold credit clock · pay on due-date from terms
V3	L3 Cash/Bank-Out	Payment already fired (pre-pay) · receive validates qty
V4	L3 Cash/Bank-Out	One-off · pay at receive · no recurring relationship
V5	L3 Cash/Bank-Out	No PO · bill arrives monthly · pay-on-due
SUPPLIER	Vendor master · L3 reconciliation	Who to pay · which contact · which payment terms apply
BRAND-SKU	Price-Intel agent	Exact SKU to scrape on Hyperpure / Blinkit / Zepto for live price
RECIPE (I-layer only)	L4 Inventory Settlement	Upstream RMs to consume + ratios; output RM to produce

WHY THIS ARCHITECTURE HOLDS